

- 1 Blood vessels must be able to withstand and maintain varying blood pressures.

There are several methods that could be used to determine the strength of a blood vessel. One method is called the circumferential tensile strength (CTS) test, as shown in Fig. 1.1.

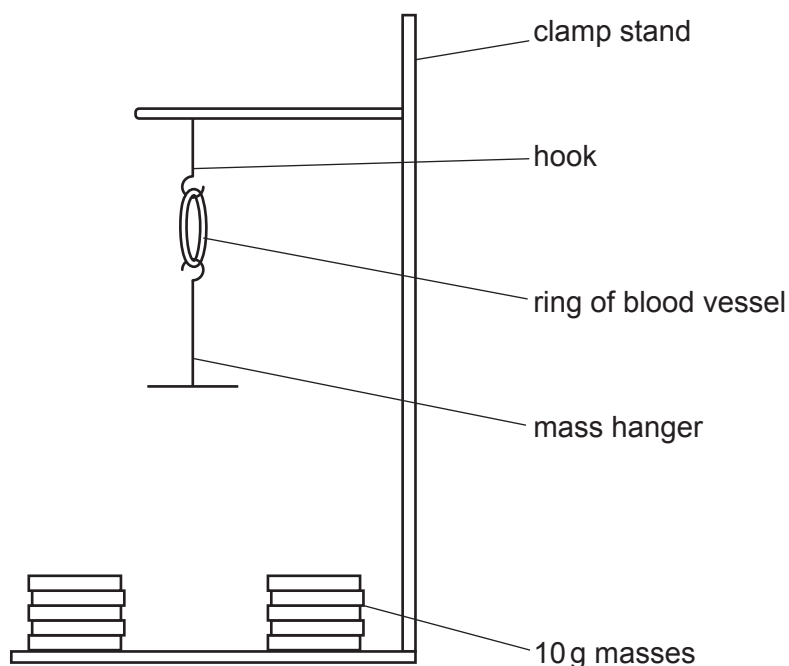


Fig. 1.1

The ring of blood vessel, shown in Fig. 1.1, was cut from a length of artery or vein that was prepared by removing surrounding tissues.

A student wanted to use the apparatus in Fig. 1.1 to determine the mass needed to break an artery and the mass needed to break a vein.

- (a) (i) Identify the independent variable **and** the dependent variable in this investigation.

independent variable

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dependent variable

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- (ii) The student was provided with lengths of a large artery, lengths of a large vein and standard laboratory apparatus.

Describe how the student could use the CTS test to determine the mass needed to break an artery and the mass needed to break a vein.

Your method should be set out in a logical order and be detailed enough to let another person follow it.

[6]

Another student investigated how the length of a ring of vein and the length of a ring of artery increases as more mass is added.

The student used the apparatus shown in Fig. 1.2 with the ring of vein.

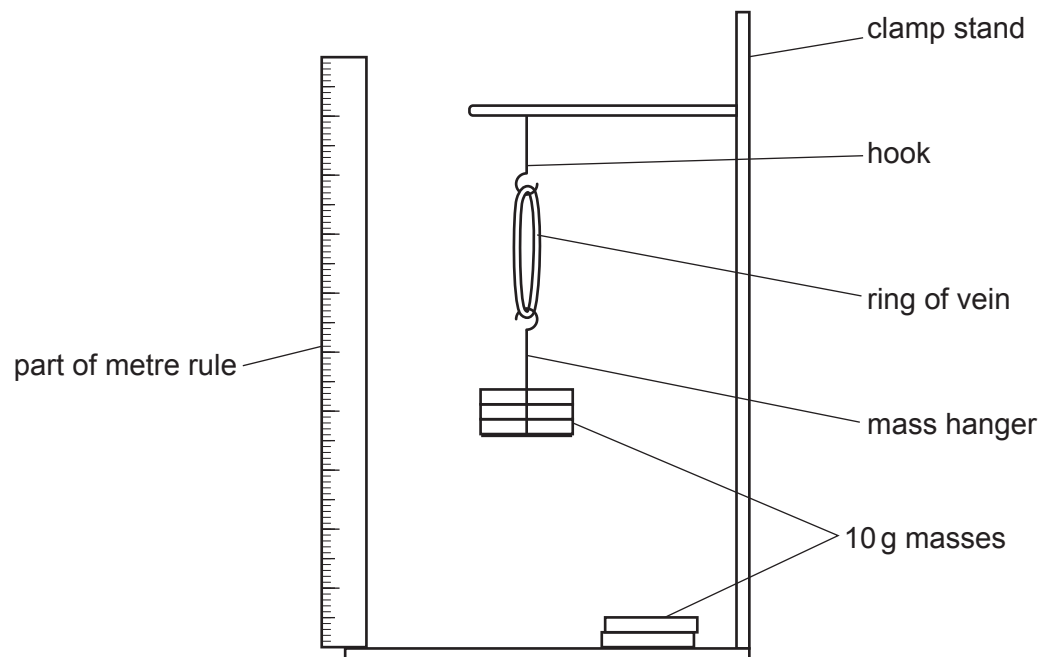


Fig. 1.2

The results for the ring of vein are shown in Table 1.1.

Table 1.1

mass added / g	length of ring of vein / mm	percentage increase in length of ring of vein
0	21	0
10	36	
20	38	81
30	40	90
40	41	
50	41	

- (b) (i) Complete Table 1.1 by calculating the percentage increase in length of the ring of vein for 10g, 40g and 50g. [2]

- (ii) Explain why the student calculated the percentage increase in length of the ring of vein.

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..... [1]

- (iii) On the grid in Fig. 1.3, plot a graph of the data shown in Table 1.1.

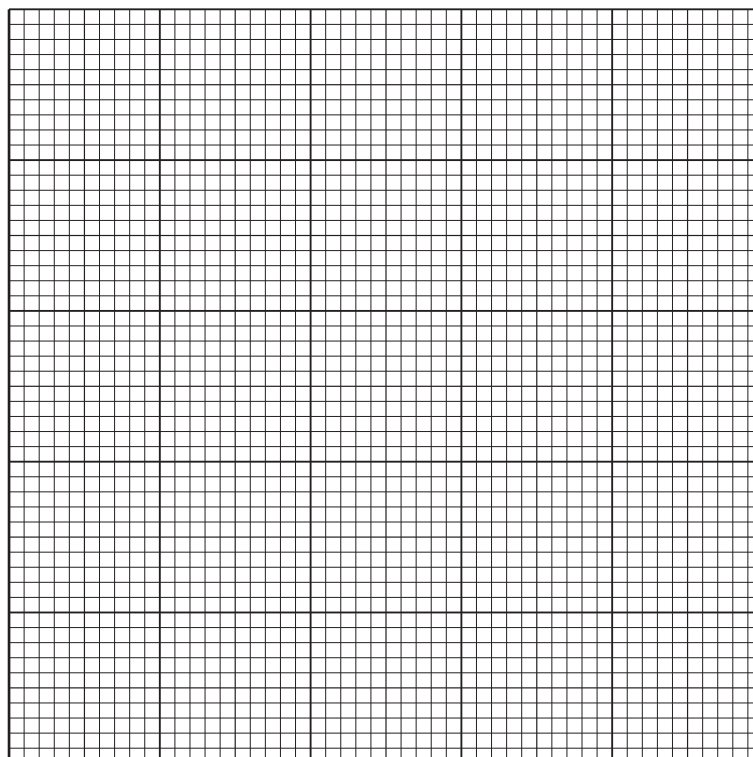


Fig. 1.3

[3]

- (iv) The student carried out the same investigation using a muscular artery instead of a vein.

Predict the shape of the curve you would expect for the muscular artery.

On Fig. 1.3:

- sketch the curve you predicted for the muscular artery
- label this curve with the word **artery**.

[2]

(v) Explain the shape of the curve you sketched for the muscular artery in Fig. 1.3.

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..... [2]

(vi) Another student suggested that the experiment should be repeated with more rings from the same blood vessels.

Suggest **two other** ways the student could modify the method to improve the quality of the results.

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..... [2]

[Total: 19]